Maya R. Patel

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EDUCATION

The University of Texas at Austin — Austin, TX
Bachelor of Science, Computer Science; Minor in Applied Statistics — Aug. 2022 - May 2026

GPA: 3.82/4.00 | **Honors:** University Honors (6 semesters), CNS Dean's Scholars Program
Relevant Coursework: Machine Learning, Data Mining, Distributed Systems, Database Management, Probability, Algorithms, Natural Language Processing, Cloud Computing, Software Engineering

PROFESSIONAL EXPERIENCE

Machine Learning Engineering Intern — Austin, TX
Bluebonnet Health Analytics — May 2025 - Aug. 2025

- Built a Python and PyTorch pipeline to classify clinical referral documents across **18 categories**, improving macro F1 from **0.71 to 0.86** through class-balanced sampling, feature ablations, and error-driven labeling.
- Designed an asynchronous OCR and inference service with FastAPI, Celery, Redis, and Docker that processed **65,000+ pages per week** while reducing median end-to-end latency from 9.4 seconds to **3.1 seconds**.
- Implemented drift monitoring for 27 structured and text-derived features, creating automated alerts that helped the data team identify a vendor format change within **24 hours** instead of several weeks.
- Partnered with product and compliance teams to document model limitations, define human-review thresholds, and launch a staged pilot used by **42 operations specialists** across three clinics.

Undergraduate Research Assistant — Austin, TX
UT Austin Responsible AI Laboratory — Jan. 2024 - May 2026

- Evaluated fairness and calibration of large-language-model classifiers on **120,000** public benefits questions, comparing demographic parity, equalized odds, expected calibration error, and abstention strategies.
- Developed reproducible experiments with Python, Hugging Face, Weights & Biases, and Slurm; reduced full benchmark runtime by **38%** using batched tokenization, cached embeddings, and mixed-precision inference.
- Co-authored a workshop paper accepted to the **2025 ACM Conference on Fairness, Accountability, and Transparency student track**; created visualizations and drafted the evaluation methodology section.
- Mentored two first-year researchers on Git workflows, experiment design, statistical significance testing, and responsible handling of sensitive attributes.

Data Engineering Intern — Remote
CivicSignal Labs — Sep. 2024 - Dec. 2024

- Created Airflow DAGs and dbt models that consolidated transit, weather, and service-alert feeds into BigQuery, increasing successful daily loads from **91% to 99.6%**.
- Added schema tests, freshness checks, lineage documentation, and incident runbooks for **34 production tables**, reducing recurring data-quality tickets by **31%**.
- Built a geospatial aggregation API in Python and PostGIS that served neighborhood-level reliability metrics with p95 response times below **180 ms**.

SELECTED PROJECTS

MetroPulse - Transit Delay Forecasting | *Python, LightGBM, FastAPI, React, PostGIS, Docker* — Jan. 2025 - Apr. 2025

- Trained gradient-boosted models on **14 million** vehicle observations to predict 15-minute arrival delays, achieving a **22% lower MAE** than schedule-based baselines.
- Deployed a containerized inference API and interactive map supporting route, stop, time-of-day, and weather filters for a 600-user campus transportation pilot.

ADDITIONAL PROJECTS

PaperTrail - Document Intelligence Platform | *Python, Transformers, OpenCV, PostgreSQL, Next.js, AWS* Aug. 2025 - Dec. 2025

- Developed a multi-stage extraction pipeline for invoices and receipts using image quality checks, OCR confidence scoring, table detection, and field-level validation; reached **93.4% exact-match accuracy** on 1,800 held-out documents.
- Added queue-based processing with SQS and Lambda, idempotency keys, retry policies, and structured logs; sustained **40 concurrent uploads** without duplicate records or job loss.
- Created a Next.js review interface that highlighted low-confidence fields and captured corrections for active-learning experiments, cutting average review time by **46%**.

PantryCast - Food Pantry Demand Planner | *Python, Prophet, XGBoost, Streamlit, DuckDB* Feb. 2024 - May 2024

- Forecasted weekly demand for 23 food categories using three years of anonymized distribution records, holidays, weather, and academic-calendar features; reduced weighted forecast error by **19%**.
- Built scenario controls for donation variability and safety-stock targets, enabling volunteers to compare expected shortages under multiple budget assumptions.
- Presented findings to a local nonprofit board and delivered a documented handoff package with data dictionary, retraining steps, and model-monitoring guidance.

Campus Research Search | *TypeScript, Node.js, Elasticsearch, OpenAI-compatible embeddings* Sep. 2023 - Dec. 2023

- Indexed 7,500 faculty publications and laboratory pages with hybrid keyword and vector retrieval, improving top-five relevance by **28 percentage points** over keyword search alone.
- Implemented query logging, relevance feedback, rate limiting, and evaluation scripts for nDCG@10 and mean reciprocal rank.

LEADERSHIP, RESEARCH & ACTIVITIES

Teaching Assistant - Elements of Data Science Austin, TX
UT Austin Department of Statistics and Data Sciences Aug. 2024 - May 2026

- Led two weekly lab sections for 58 students covering pandas, visualization, hypothesis testing, regression, and reproducible notebooks; maintained a median course-evaluation score of **4.8/5**.
- Created autograded assignments and debugging guides that reduced repeated office-hour questions by **27%**.

Data Science Lead Austin, TX
Women in Computing at UT Austin May 2024 - May 2025

- Organized six technical workshops on SQL, machine learning, and responsible AI for 180+ attendees and coordinated 14 volunteer instructors.
- Led a five-person team that analyzed event attendance and membership retention, informing a revised mentorship program that increased returning participation by **21%**.

CERTIFICATIONS & RECOGNITION

• **AWS Certified Cloud Practitioner** - Amazon Web Services, issued Nov. 2025
• **Databricks Lakehouse Fundamentals** - Databricks, issued Jul. 2025
• **HackTX 2024 - Best Use of Public Data** for a heat-risk resource-mapping application

TECHNICAL SKILLS

Programming: Python, SQL, TypeScript, Java, C++, Bash, R
Machine Learning & Data: PyTorch, scikit-learn, XGBoost, LightGBM, Hugging Face, pandas, NumPy, dbt, Airflow, Spark, DuckDB
Cloud & Infrastructure: AWS (S3, Lambda, SQS, ECS), Google Cloud (BigQuery), Docker, Kubernetes, Terraform, GitHub Actions, Linux
Web & Databases: FastAPI, Flask, Node.js, React, Next.js, PostgreSQL, PostGIS, Redis, Elasticsearch
Methods: Experiment design, model evaluation, feature engineering, data quality testing, API design, observability, technical writing

ADDITIONAL INFORMATION

Work Authorization: Authorized to work in the United States without current or future sponsorship
Languages: English (native), Gujarati (conversational), Spanish (intermediate)
Interests: Urban mobility, public-interest technology, long-distance running, community food systems